Laboratory Manual Adaptive Control & Reinforcement Learning

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9th Semester

Outline

PART I (coupled DC motors)

- Lab I.1: Modeling and identification of a pair of mechanically coupled permanent magnet DC motors.
- Lab I.2: Adaptive velocity control of the coupled DC motors with th
- Lab I.3: Velocity control of the coupled DC motors with MRAC (scala
- Lab I.4: Velocity control of the coupled DC motors with ADI (scalar

- Lab I.5: Robust adaptive velocity control of the coupled DC motors (scalar case).
- Lab I.6: Robust adaptive angle control of the coupled DC motors (vector case).

PART II (coupled water tanks)

- Lab II.1: Modeling, linearization and identification of two coupled water tanks.
- Lab II.2: Adaptive control of the coupled tanks with the MIT rule.
- Lab II.3: Adaptive feedback linearization control of the coupled tanks.
- Lab II.4: Adaptive backstepping control of the coupled tanks.

LAB I.1 - Description

Model the coupled DC motors system with a first order transfer function (pole and gain) and identify the parameters using step inputs.

- ✓ Measure for various positive and negative voltage inputs the steady state velocity and build a table with input (pwm) and output (ω). Estimate the gain.
- ✓ Measure for various positive and negative voltage inputs the rise time and build a table with input (pwm) and rise time (τ). Estimate the pole.
- ✓ Verify the model using various input signals (step, slow sinusoidal, ramp).
- ✓ Design and implement a P-controller and a PI-controller based on the estimated model (use the identified parameters to tune the control gains).
- ✓ Vary the potentiometer connected to the second motor and see how it affects the dynamics in step responses.

LAB I.2 - Description

Design and implement an adaptive control law for the coupled DC motors.

- ✓Design a linear reference model (a model that meets the specifications provided by the designer – how would the designer to operate the coupled DC motor system). Use either a potentiometer to provide commands or via code.
- ✓Design an adaptive controller based on the MIT rule to follow the reference model. Adopt the appropriate model. Use both step reference signals of various levels as well as sinusoidal reference signals.
- ✓Try to extract conditions for the stability of the closed loop system.

LAB I.3 - Description

Design and implement an adaptive control law for the coupled DC motors.

- ✓ Design a linear reference model (a model that meets the specifications provided by the designer – how would the designer to operate the coupled DC motor system). Use either a potentiometer to provide commands or via code.
- ✓ Design an adaptive controller based on the scalar MRAC approach to follow the reference model. Adopt the appropriate model. Use both step reference signals of various levels as well as slow sinusoidal reference signals.
- ✓ For simple step reference leave the adaptive controller to operate for a long period and see if instability occurs.
- ✓ Try slow sinusoidal reference signals and see if the PE condition is met based on the identified model and the reference one.

LAB I.4 - Description

Design and implement an adaptive control law for the coupled DC motors.

- ✓ Design a linear reference model (a model that meets the specifications provided by the designer – how would the designer to operate the coupled DC motor system). Use either a potentiometer to provide commands or via code.
- ✓ Design an adaptive controller based on the scalar ADI approach to follow the reference model. Adopt the appropriate model. Use both step reference signals of various levels as well as slow sinusoidal reference signals.
- ✓ For simple step reference leave the adaptive controller to operate for a long period and see if instability occurs.
- ✓ Try slow sinusoidal reference signals and see if the PE condition is met based on the identified model and the reference one.

LAB I.5 - Description

Design and implement robust adaptive control laws for the coupled DC motors.

- ✓ Design a linear reference model (a model that meets the specifications provided by the designer – how would the designer to operate the coupled DC motor system). Use either a potentiometer to provide commands or via code.
- ✓ Design robust adaptive controllers based on the scalar MRAC and ADI approaches to follow the reference model. Adopt a linearly parameterized approximation model of the uncertainties and appropriate robustifying techniques (deadzone, σ-modification, e-modification, projection). Use both step reference signals of various levels as well as slow sinusoidal reference signals.
- ✓ For simple step reference leave the adaptive controller to operate for a long period and see if instability occurs.
- ✓ Try slow sinusoidal reference signals and see if the PE condition is met based on the identified model and the reference one.

LAB I.6 - Description

Design and implement robust adaptive angle control laws for the coupled DC motors.

- ✓ Design a linear reference angle model (a model that meets the specifications provided by the designer – how would the designer to operate the coupled DC motor system). Use either a potentiometer to provide commands or via code.
- ✓ Design a robust adaptive controller based on the vector MRAC approach to follow the reference angle model. Adopt a linearly parameterized approximation model of the uncertainties and appropriate robustifying techniques (deadzone, σ-modification, e-modification, projection). Use step reference signals of various levels as well as slow sinusoidal reference signals.
- ✓ For simple step reference leave the adaptive controller to operate for a long period and see if instability occurs.
- ✓ Try slow sinusoidal reference signals and see if the PE condition is met based on the identified model and the reference one.

LAB II.1 - Description

Model the two coupled tanks system with a second order nonlinear system.

- ✓Measure the various system parameters and identify an equilibrium point for the tank levels and the input water flow. Linearize around that point. Implement and find the actual equilibrium.
- ✓Measure for various positive and negative input water flows the response and estimate the linearized model.
- ✓Verify the model using various input signals (step, slow sinusoidal, ramp).
- ✓Design and implement a P-controller, a PI-controller and a PID-controller based on the estimated model (use the identified parameters to tune the control gains).

LAB II.2 - Description

Design and implement an adaptive control law for the coupled tanks system.

- ✓Design a second order linear reference model (a model that meets the specifications provided by the designer – how would the designer to operate the coupled tanks system). Use either a potentiometer to provide commands or via code.
- ✓Design an adaptive controller based on the MIT rule to follow the reference model. Adopt the appropriate control model. Use both step reference signals of various levels as well as sinusoidal reference signals.
- ✓Try to extract conditions for the stability of the closed loop system.

LAB II.3 - Description

Design and implement an adaptive feedback linearization control law for the coupled tanks system.

- ✓ Design a linear reference model (a model that meets the specifications provided by the designer – how would the designer to operate the coupled tanks system). Use either a potentiometer to provide commands or via code.
- ✓ Design an adaptive controller based on the feedback linearization technique to follow the reference model. For simple step reference leave the adaptive controller to operate for a long period and see if instability occurs. Tune the gains appropriately.
- ✓ Adopt a linearly parameterized approximation model of the uncertainties and appropriate robustifying techniques (deadzone, σ-modification, e-modification, projection). Use both step reference signals of various levels as well as slow sinusoidal reference signals.
- ✓ Try slow sinusoidal reference signals and see if the PE condition is met based on the identified model and the reference one.

LAB II.4 - Description

Design and implement an adaptive backstepping control law for the coupled tanks system.

- ✓ Design a linear reference model (a model that meets the specifications provided by the designer – how would the designer to operate the coupled DC motor system). Use either a potentiometer to provide commands or via code.
- ✓ Design an adaptive controller based on the backstepping technique to follow the reference model. For simple step reference leave the adaptive controller to operate for a long period and see if instability occurs. Tune the gains appropriately.
- ✓ Adopt a robustifying technique (deadzone, σ-modification, e-modification, projection). Use both step reference signals of various levels as well as slow sinusoidal reference signals.
- ✓ Try slow sinusoidal reference signals and see if the PE condition is met based on the identified model and the reference one.